COMPANION TO THE PRD · GAP ANALYSIS

# MVP Gap Analysis

Post-Operative Remote Patient Monitoring: where the build stands against the PRD, and what stands between it and a first real patient.

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Version 1.1 · For product, clinical and engineering review

Assesses the main branch at commit 274f140 against PRD v0.1 (14 September 2026)

28 September 2026

SECTION 1

# Summary

The product is a convincing demo and not yet an MVP. On seeded data it does what the PRD says matters most: a nurse queue that sorts itself by clinical risk. It is not ready for a real patient, even at one small hospital, because a critical finding pages nobody, patients cannot sign in with a passcode and clinicians have no second factor, and the system has never run anywhere but a developer's machine.

The PRD itself separates a *clickable demo on seeded data* from a *feature-complete deployable MVP*, and puts about four months between the demo and the first real patient. The build is clearly past the first milestone and short of the second.

| 20 | 7 | 6 | 5 |
|---|---|---|---|
| of 33 functional requirements met | functional requirements partly met | functional requirements not met | of 7 blockers still open before a first real patient |

## What is genuinely strong

- **The core product bet is built.** The risk-sorted queue, the rules engine to its specification, day-indexed and trend-aware scoring, versioned rulesets stamped on every decision, missed check-in detection, alert de-duplication, escalation timing and mandatory outcomes all work.
- **It already serves several hospitals.** Each hospital's data is kept apart by the database itself, and a platform admin area manages hospitals and rulesets. The PRD deferred this (NG-3); it is exactly what selling to small and mid-tier hospitals needs.
- **Enrolment captures a discharge baseline and versioned consent,** and the WhatsApp reminder, signed check-in link and patient check-in form work end to end.
- **Surgeons can change thresholds without a code change** (goal G-5), through ruleset drafts and approval.
- **It is well tested:** 298 backend and 95 frontend automated tests pass.

> **BOTTOM LINE**
>
> Describe it today as a working product demo, or a pilot-ready prototype. It becomes an MVP when the five open blockers in Section 3 are closed. Four are software and infrastructure; the fifth is clinical and operational work that the PRD says sets the date.

## Changes in version 1.1

Version 1.0 assessed commit `44d8caf`. Commit `274f140` has since closed blockers 2 and 3 and part of blocker 4:

- The check-in simulator is off unless switched on for demo data, so patient records are no longer reachable without signing in. A check-in needs the patient's WhatsApp link.
- Each hospital sets its own emergency number, which the patient's call button dials. Until it is set, the button dials emergency services on 112.
- Check-in links keep their token out of every server's logs.

SECTION 2

# How this was assessed

The yardstick is the PRD (version 0.1, 14 September 2026): its goals, its 33 functional and 15 non-functional requirements, and its delivery plan. The build assessed is the `main` branch at commit `274f140` (28 September 2026).

Each requirement's status comes from reading the code and running the product, not from documentation. The app was started against its Neon database and driven in a browser: clinician sign-in, the risk queue, hospital WhatsApp settings, and a patient check-in opened from a WhatsApp link through to submission. Both automated test suites were run.

## Status labels

| Label | Meaning |
|---|---|
| **Met** | Built and working as the PRD describes. |
| **Partial** | Some of the requirement is built; the note says what is missing. |
| **Not met** | Not built. |
| **Not measured** | Cannot be judged from code: needs a load test, device test, audit or pilot. |

## Limits of this assessment

- No load, latency or availability testing, and no testing on real Android handsets.
- No accessibility audit and no usability testing with patients in the 55 to 75 age band.
- No clinical validation: whether the thresholds flag the right patients is exactly what a pilot must establish.
- No legal review. Section 6 flags regulatory questions; it does not answer them.

SECTION 3

# Blockers before a first real patient

Every open item below must be closed before the product is used with a real patient, even at a single friendly clinic. They are ordered by the harm they could do. Items closed since version 1.0 stay in the list, marked closed, so the numbers still match earlier references.

### 1. Nobody is paged when a patient deteriorates OPEN

Critical and emergency alerts appear on the dashboard and escalate on time, but notifications can only be switched off or written to a log. No SMS, push or call reaches the on-call nurse, the backup nurse or the surgeon. The dashboard itself warns that nobody is paged.

| | |
|---|---|
| **Why it matters** | FR-23, FR-25, FR-33, NFR-2 and goal G-3: no critical finding goes unacknowledged. |
| **What closes it** | Add an SMS (and ideally push) delivery provider to the existing notification queue, and a roster of who is on call. |
| **Where** | `backend/app/core/config.py (NOTIFICATION_PROVIDER)` |
| **Effort** | PRD: SMS integration, 2 engineer-days; the alert and escalation logic already exists. |

### 2. Patient records were reachable without signing in CLOSED

The check-in simulator's routes needed no login. One listed the name and hospital number of every active patient in every hospital; others built any patient's form and accepted a check-in for any patient.

| | |
|---|---|
| **Why it matters** | Patient privacy (DPDP) and data integrity: anyone with the address could read names or file clinical answers. |
| **How it was closed** | Closed in 274f140. The simulator answers only when CHECKIN_SIMULATOR_ENABLED is set, for demos on seeded data, and is off by default. Without it a check-in needs the patient's WhatsApp link, and the check-in page without one tells the patient to use it. |
| **Where** | `backend/app/api/v1/checkin.py, backend/app/core/config.py` |
| **Effort** | Done. |

### 3. The emergency call button dialled a placeholder number CLOSED

When a patient reported an emergency symptom, the check-in's Call the ward button dialled the same hard-coded number, +91 422 200 0000, for every hospital.

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|---|---|
| **Why it matters** | FR-22 and the PRD's mitigation for a missed complication: emergency guidance that does not depend on anyone acknowledging an alert. |
| **How it was closed** | Closed in 274f140. A superadmin sets the hospital's emergency number in hospital settings; the button dials it, and message wording can include it as {hospital_phone}. Until it is set, the button dials emergency services on 112. That fallback wording still needs clinical review. |
| **Where** | `db/migrations/014_hospital_emergency_phone.sql, frontend/src/app/checkin/page.tsx` |
| **Effort** | Done. |

### 4. Patient sign-in and reminders are unfinished PARTLY CLOSED

There is no one-time passcode and no SMS fallback; patients reach the form only through the daily WhatsApp link. WhatsApp has so far only run in development mode: each hospital's message templates must be approved by Meta before a real reminder can go out. Since 274f140, link tokens no longer reach server logs.

| | |
|---|---|
| **Why it matters** | FR-5, FR-6, NFR-13; adherence is a high-severity PRD risk. |
| **What closes it** | Patient OTP, SMS fallback for reminders, and Meta template approval for the pilot hospital. |
| **Where** | `backend/app/services/patient_messages.py, backend/app/api/v1/checkin.py` |
| **Effort** | PRD: patient OTP and clinician auth, 6 engineer-days; reminders and SMS fallback, 4 (WhatsApp covers part). |

### 5. Clinician access has no second factor and no access log OPEN

Clinicians sign in with a password only, and nothing records who opened which patient's record. Evaluations, acknowledgements and outcomes are recorded with who and when.

| | |
|---|---|
| **Why it matters** | NFR-13 (second factor) and NFR-9 (auditability); expected by any hospital's information-security review. |
| **What closes it** | Add a second factor for clinicians and a log of record access. |
| **Where** | `backend/app/api/v1/auth.py` |
| **Effort** | Included in the PRD's 6 engineer-days for authentication; the access log is not estimated. |

### 6. There is no production environment OPEN

The product has only run on a developer's machine. There is no continuous integration, container or hosting set-up, HTTPS, monitoring, alerting on the system itself, or rehearsed backup restore.

| | |
|---|---|
| **Why it matters** | NFR-3 (99.5% availability of check-in and alerting), NFR-11 (encryption in transit). |
| **What closes it** | Hosting with HTTPS, CI running both test suites, uptime and error monitoring, and a restore drill. |
| **Where** | `Repository root: no Dockerfile, no .github workflows` |
| **Effort** | PRD: infrastructure, CI, storage and monitoring, 5 engineer-days. |

### 7. Clinical and operational prerequisites OPEN

None of these is software, and the PRD says they, not engineering, set the date: surgeon sign-off on every threshold; a named on-call nurse with a committed coverage window; two rounds of usability testing with patients aged 55 to 75; a consent process and a 72-hour breach process; and a regulatory opinion (Section 6).

| | |
|---|---|
| **Why it matters** | PRD Section 8: roughly four months to the first real patient, with these on the critical path. |
| **What closes it** | Start all of them now, in parallel with the software work. |
| **Where** | PRD Sections 3, 6 and 8 |
| **Effort** | Not engineering effort; calendar time. |

SECTION 4

# Requirement by requirement

Every functional and non-functional requirement in the PRD, with its status in the build. Requirement text is shortened; the PRD has the full wording.

## 4.1 Functional requirements

20 met, 7 partial, 6 not met.

| ID | Requirement (PRD, shortened) | Status | Notes |
|---|---|---|---|
| FR-1 | Enrol with name, phone, language, surgery and discharge dates | Met | Enrolment screen and API. |
| FR-2 | Capture the discharge baseline | Met | Baseline readings stored as the comparison anchor. |
| FR-3 | Explicit, versioned consent with timestamp and text version | Met | Consent recorded per purpose with form version and privacy law. |
| FR-4 | Assign an approved ruleset; surgeon may override per patient | Partial | Ruleset assigned. No per-patient threshold overrides. |
| FR-5 | Enrolment link by SMS; one-time passcode sign-in | Not met | No SMS and no passcode. Patients use the daily WhatsApp link; the enrolment welcome message is never sent. |
| FR-6 | Daily reminder at a local time; SMS fallback | Partial | WhatsApp reminder inside the patient's check-in window. No SMS fallback; Meta templates not yet approved. |
| FR-7 | Form composed server-side from the post-op day | Met | Only the day's questions are sent. |
| FR-8 | New questions marked as new on first appearance | Not met | |
| FR-9 | Binary questions share polarity: yes is concerning | Met | Question wording follows the rule. |
| FR-10 | Binary questions default to unanswered | Met | Review step shows "Not answered". |
| FR-11 | Capture, upload and compare wound photographs | Partial | Photo is previewed on the phone only; not uploaded or stored. |
| FR-12 | Offline submissions queued and sent later | Partial | Server keeps the device's capture time and rejects replays. No queue on the phone. |
| FR-13 | Missing fields recorded as unknown, never normal | Met | Unanswered questions are omitted and scored as unknown. |
| FR-14 | Patient never sees a score, tier or alert status | Met | |
| FR-15 | Cadence change announced on the check-in before it | Not met | |
| FR-16 | Missed check-in raises its own status | Met | Hourly sweep judges yesterday in the hospital's timezone. |
| FR-17 | Every submission gets one tier | Met | Normal, review, critical, emergency; missed is separate. |

| ID | Requirement (PRD, shortened) | Status | Notes |
|---|---|---|---|
| FR-18 | Thresholds, day-indexed ranges and trend from own history | Met | Rules engine built to its specification. |
| FR-19 | A single maximum-severity signal escalates on its own | Met | |
| FR-20 | Moderate findings may escalate on the weighted total | Met | |
| FR-21 | Record ruleset version, signals and ranked contribution | Met | Stored with every evaluation. |
| FR-22 | Breathlessness or chest pain: emergency tier, guidance, team told | Partial | Tier and on-screen guidance work, and the call button dials the hospital's number or 112. Nobody is paged (blocker 1). |
| FR-23 | Critical and emergency alerts page the on-call nurse | Not met | Blocker 1. |
| FR-24 | Review tier queues without interrupting | Met | |
| FR-25 | Unacknowledged critical alert escalates after a window | Partial | Escalation is marked on time; the notification is never delivered. |
| FR-26 | Repeat alerts suppressed for 24 hours unless the tier rises | Met | |
| FR-27 | Every alert needs a recorded outcome | Met | |
| FR-28 | Queue ordered by tier then age, decided by the server | Met | |
| FR-29 | Reasons in clinical language; no numeric score | Met | |
| FR-30 | Trends against baseline, plus the wound photo series | Partial | Trends shown. No photos (FR-11). |
| FR-31 | Opening an alert shows the question script for its rule | Not met | |
| FR-32 | Resolution requires one of four outcomes | Met | Reassured, plan adjusted, same-day review, emergency. |
| FR-33 | Emergency outcome notifies the surgical team | Not met | Depends on blocker 1. |

## 4.2 Non-functional requirements

1 met, 4 partial, 6 not met, 4 not measured.

| ID | Requirement (PRD, shortened) | Status | Notes |
|---|---|---|---|
| NFR-1 | Evaluation within 2 s at the 95th percentile | Not measured | Scored in the same request; not load-tested. |
| NFR-2 | Critical alert dispatched within 60 s | Not met | Nothing is dispatched (blocker 1). |
| NFR-3 | 99.5% monthly availability | Not met | No production environment (blocker 6). |
| NFR-4 | Up to 500 concurrently enrolled patients | Not measured | Design matches the PRD's no-broker assumption. |

| ID | Requirement (PRD, shortened) | Status | Notes |
|---|---|---|---|
| NFR-5 | Accessibility: 16 px text, 44 px targets, WCAG AA | Not measured | Large touch targets; not audited. |
| NFR-6 | English, Tamil and Hindi at launch | Not met | Screens are English only; Tamil and Hindi message wording is unwritten. |
| NFR-7 | Android 9+, current browsers, works on 3G | Not measured | Not tested on devices. |
| NFR-8 | Seven days of offline submissions kept | Not met | No queue on the phone. |
| NFR-9 | Every evaluation, alert, notification and outcome audited | Partial | Recorded with who and when; no log of record access. |
| NFR-10 | Re-evaluation reproduces the original result | Met | Deterministic evaluator; fixture patients check it. |
| NFR-11 | Encryption in transit and at rest | Partial | Database connection encrypted and stored encrypted by Neon; no HTTPS deployment yet. |
| NFR-12 | Photos only via short-lived signed URLs | Not met | No photo storage yet. |
| NFR-13 | Patient passcode; clinician second factor; roles | Partial | Roles and per-hospital isolation work. No passcode, no second factor. |
| NFR-14 | Versioned, purpose-limited consent; withdrawal supported | Partial | Versioned, purpose-coded consent captured. No withdrawal flow. |
| NFR-15 | Breach process: notify within 72 hours | Not met | No documented process. |

## 4.3 Goals

| Goal | Status | Notes |
|---|---|---|
| G-1 | Not measured | Detect complications earlier than the follow-up. The capability is built; only a pilot can show it works. |
| G-2 | Met | One queue ordered by clinical urgency; the nurse makes no triage decision. |
| G-3 | Not met | No critical finding unacknowledged: escalation fires, but nobody is paged (blocker 1). |
| G-4 | Not measured | Low patient burden: 12 questions in 5 short steps on day 21, but no usability testing yet. |
| G-5 | Met | Surgeons adjust thresholds through ruleset drafts and approval, without a code change. |

SECTION 5

# Beyond the blockers

## 5.1 To meet the MVP as the PRD specifies

Needed for the product the PRD describes, once real patients are safe to enrol.

- **Wound photographs**: upload, storage, signed links, previous-day comparison and the clinician's photo series (FR-11, FR-30, NFR-12). The PRD's main infection signal for days 0 to 14. PRD estimate: 4 engineer-days.
- **Offline queue on the phone** for up to seven days (FR-12, NFR-8). The server side is ready. PRD estimate: 5 engineer-days.
- **Tamil and Hindi**, professionally translated, for screens and messages (NFR-6).
- **A question script** for the nurse when opening an alert (FR-31).
- **Per-patient threshold overrides** by the surgeon at discharge (FR-4).
- **New-question markers and cadence-change notices** on the check-in (FR-8, FR-15).
- **Consent withdrawal** (NFR-14).
- **A signed-in screen for nurses to enter a check-in taken by phone.** The database already accepts a nurse as the source; nothing writes it. The PRD's recommended validation run (Section 7) depends on it.
- **Measurement** of latency, load, accessibility and device support (NFR-1, NFR-4, NFR-5, NFR-7).

## 5.2 To sell to more hospitals

The PRD scopes the MVP to one clinic and one procedure. Selling to small and mid-tier hospitals adds:

- **Onboarding.** A hospital created in the platform admin area starts with no WhatsApp wording, so its reminders are recorded as failed until a superadmin writes every message by hand. Each hospital also needs its own WhatsApp Business verification and Meta template approval, and a superadmin to set its emergency number.
- **More procedures.** Only the knee replacement ruleset exists. The architecture is procedure-agnostic, but each new procedure needs a surgeon-approved ruleset.
- **Already in place:** per-hospital data isolation, per-hospital WhatsApp accounts, and a platform admin for hospitals and rulesets.

SECTION 6

# Regulatory and compliance

**Data protection (DPDP).** As the PRD notes, the DPDP Rules were notified on 13 November 2025, the Consent Manager framework takes effect on 13 November 2026, and the principal substantive provisions from 13 May 2027. Built: versioned, purpose-coded consent at enrolment, and per-hospital data isolation. Missing: consent withdrawal, a 72-hour breach notification process, and a log of who accessed which record.

**Medical device regulation.** Software that sorts patients by clinical risk may be regulated as a medical device under India's Medical Device Rules, 2017. Get a regulatory opinion before any hospital pilot; the answer may change what has to be documented and validated.

SECTION 7

# Recommended path

| Stage | Milestone | What it needs |
|---|---|---|
| Stage 0 | **Demo on seeded data** | Ready today. The repository's run-simple-eras skill starts the app, resets the demo patients and drives a scripted walkthrough. |
| Stage 1 | **Validation run (the PRD's recommendation)** | Rules engine and dashboard fed by nurses entering phone check-ins at one friendly clinic for two weeks, to prove the queue flags the right patients before the patient app matters. Needs blockers 5 and 6, surgeon-approved thresholds, and the nurse entry screen. |
| Stage 2 | **First real patients on the app** | The five open blockers closed: 1, 4, 5, 6 and 7. |
| Stage 3 | **MVP as the PRD specifies** | Section 5.1 complete. |
| Stage 4 | **More hospitals** | Section 5.2 complete, plus the regulatory position settled. |

## Remaining engineering effort

The PRD's own estimates for the workstreams that are still open add up to about 26 engineer-days: patient passcode and clinician authentication (6), SMS integration (2), reminders and SMS fallback (up to 4, as WhatsApp covers part), photo capture and upload (4), offline queue (5), and infrastructure, CI, storage and monitoring (5). That excludes work the PRD did not estimate: the record-access log, translation, question scripts, per-patient overrides and the nurse entry screen.

> **THE DATE IS NOT SET BY ENGINEERING**
>
> As the PRD says, the critical path runs through surgeon sign-off, rostering a nurse to answer alerts, usability testing and the consent and breach processes. Start them now, in parallel with the software work.